

FMUSER FUQ-100B Modular Encoder-Receiver-IP Gateway IPTV Server 3-6 Card Slots HDMI-Tuner-IP MPEG TS Input SRT-HTTP-UDP-RTP-RTSP-HLS-RTMP Protocol Conversion Hotel Hospital Community IPTV



Multiple Inputs (HDMI+Tuner+IP over Multi-Protocol+TS file) + Multi-Protocol Conversion + IPTV Server



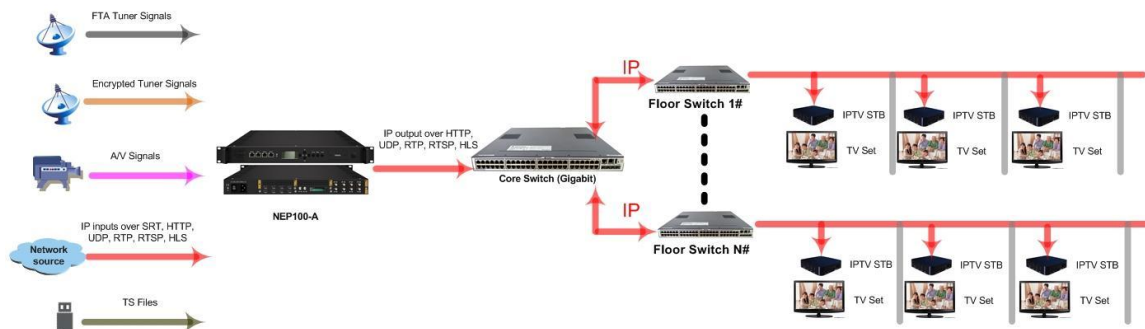
1U Chassis with 3 card slots



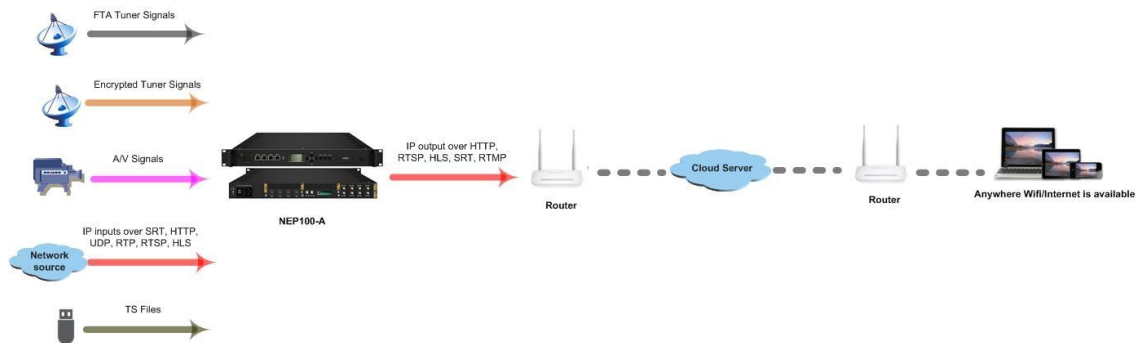
3U Chassis with 6 card slots

Application Example:

IPTV System as a Server



Protocol Conversion as a Streamer



Product overview

The FUQ-100B integrates Encoder/Receiver, IP Gateway, and IPTV Server functions, supporting inputs like HDMI, tuner signals, IP streams, and TS files. It converts protocols including SRT, HTTP, UDP, RTP, RTSP, HLS, and RTMP for reliable transmission. This modular device comes in 1U or 3U rack sizes and is controlled via Web GUI. Ideal for IPTV systems in hotels, hospitals, and communities.

Key Features

"Integrates Encoder/Receiver, IP Gateway, and IPTV Server in a compact modular device,

Supports up to 3 (1U) or 6 (3U) pluggable cards like encoder cards or tuner cards,

Enables direct TS files upload via Web GUI for custom channel broadcasting,

Features inter-cut for seamless switching between live programs, TS files, and images,

Incorporates IP anti-jitter technology to stabilize external IP streams"

IP Protocol Conversion (example)



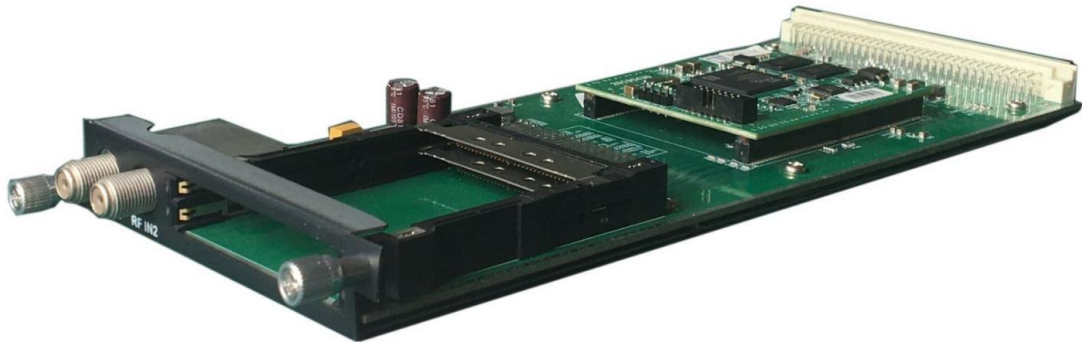
Specifications

Input	IP inputs thru ETH 1&2, GE ports over SRT, HTTP, UDP (SPTS), RTP (SPTS), RTSP (over UDP, payload: mpeg TS) and HLS	For 1U model
Input	IP inputs thru ETH 1-4 GE ports & 6-7 SFP+ ports over SRT, HTTP, UDP (SPTS), RTP (SPTS), RTSP (over UDP, payload: mpeg TS) and HLS	For 3U model
Input	TS files uploading through Web management	TS files uploading through Web management
Input	Encoder card and Tuner card etc (Please refer to the detailed card spec below)	Encoder card and Tuner card etc (Please refer to the detailed card spec below)
IP output	IP outputs thru ETH0, GE port over SRT, HTTP (Unicast), UDP (SPTS, Multicast), RTP, RTSP, HLS and RTMP (Program source should be H.264 and AAC encoding)	IP outputs thru ETH0, GE port over SRT, HTTP (Unicast), UDP (SPTS, Multicast), RTP, RTSP, HLS and RTMP (Program source should be H.264 and AAC encoding)
System	Channel switching time with FMUSER' STB: HTTP (1-3s), HLS (0.4-0.7s)	Channel switching time with FMUSER' STB: HTTP (1-3s), HLS (0.4-0.7s)
System	It is closely related with the program bitrate and protocol type etc for Max program numbers involved in protocol conversion, and the actual application shall prevail with maximum 80% CPU utilization (Please refer to Test data for reference in the end of the spec)	It is closely related with the program bitrate and protocol type etc for Max program numbers involved in protocol conversion, and the actual application shall prevail with maximum 80% CPU utilization (Please refer to Test data for reference in the end of the spec)

System	It is closely related with the program bitrate and protocol type etc for Max affordable terminal numbers in IPTV application of the STB/Android TV installed with FMUSER IPTV APK, and the actual application shall prevail with maximum 80% CPU utilization (Please refer to Test data for reference in the end of the spec)	It is closely related with the program bitrate and protocol type etc for Max affordable terminal numbers in IPTV application of the STB/Android TV installed with FMUSER IPTV APK, and the actual application shall prevail with maximum 80% CPU utilization (Please refer to Test data for reference in the end of the spec)
System	IPTV Features: Live channel, VOD, Hotel intro, Dining, Hotel service, Scenery intro, APPS, Adding scrolling caption, welcome words, pictures, advertisement, video, music etc (the features are only applicable to IP out application in the STB/Android TV installed with FMUSER IPTV APK)	IPTV Features: Live channel, VOD, Hotel intro, Dining, Hotel service, Scenery intro, APPS, Adding scrolling caption, welcome words, pictures, advertisement, video, music etc (the features are only applicable to IP out application in the STB/Android TV installed with FMUSER IPTV APK)
General	Demission (WxLxH)	482mm×464mm×44mm (1U model) 482mm×493mm×133mm (3U model)
General	Management	2 separate Web GUI (one for Cards and Gateway, the other for IPTV Server) through ETH3 (3U model through ETH5)
General	Temperature	0~45℃(operation), -20~80℃(storage)
General	Power Supply	AC100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz

Available Card Specification (more available cards later for development)

DX902A 2 Tuner Descrambling Card



Stream in:

2 Tuner input, F Type,

Stream out:

16 SPTS output over UDP/RTP

DVB-CI:2 independent common interface slots

Standard:

DVB-S/S2/S2X;

DVB-S

Input Frequency: 950-2150MHz

Symbol Rate: QPSK 1~45Msps

Signal Strength: -65~ -25dBm

FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8

DVB-S2

Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK 1~45Msps

16APSK 1~45 Msps

32APSK 1~32 Msps

FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8, 4/5, 5/6, 8/9, 9/10

DVB-S2X

Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK/16APSK 0.5~45 Msps;

8APSK/32APSK: 0.5~40Msps

FEC Demodulation:

QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18

8APSK: 5/9-L, 26/45-L

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90

32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9

Support Diseqc function

Multiplexing:

Maximum PID Remapping: 256 output pids

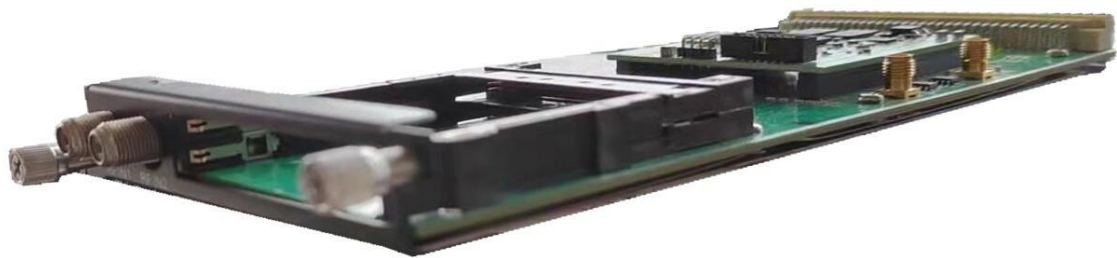
Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically

Descrambling:

CAM/CI Quantity: 2

BISS Mode: Mode 1, Mode E; 32 BISS Keys

DX942A 4 frequencies Descrambling Card



Stream in:

4 frequencies input (each RF in interface for 2 frequencies locking), F Type,

Stream out:

16 SPTS output over UDP/RTP

DVB-CI:

2 independent common interface slots

Standard: DVB-C (J.83 A/C)/J.83B/ DVB-T/DVB-T2/ISDB-T switchable

Standard: DVB-C (J.83 A/C); J.83B

Input Frequency: 60MHz~890MHz

Symbol rate: 1000~9000Ksps

Constellation: 16/32/64/128/256 QAM; 64/256 QAM for J.83B

Standard: DVB-T/T2

Frequency In: 60MHz~890MHz

Bandwidth: 5/6/7/8M bandwidth

PLP supported for DVB-T2

Standard: ISDB-T

Input Frequency: 60-890MHz

Multiplexing:

Maximum PID Remapping: 256 output pids

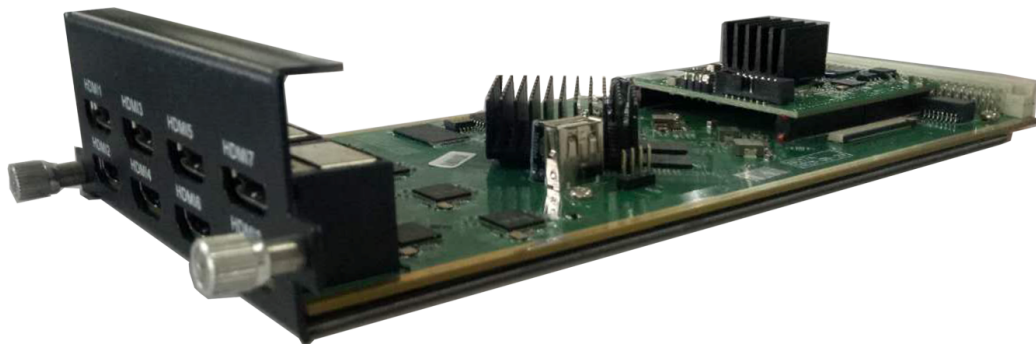
Function: PID remapping (automatically or manually), generate PSI/ SI table automatically

Descrambling:

CAM/CI Quantity: 2

BISS Mode: Mode 1, Mode E; 32 BISS Keys

DX228S 8 HDMI Encoder Card



Input: 8*HDMI (4 HDMI is available)

Output: 8*SPTS (4 SPTS if 4 HDMI) output over UDP/RTP/RTSP, Unicast/Multicast

Video Encoding:

Video format: MPEG-4 AVC/H.264

Input resolution: 1920×1080_60P, 1920×1080_60i, 1920×1080_50P, 1920×1080_50i, 1280×720_60P, 1280×720_50P, 720×576_50i, 720×480_60i,

Output resolution: 1920×1080_30P, 1920×1080_25P, 1280×720_30P, 1280×720_25P, 720×576_25P, 720×480_30P,

GOP structure: IP...P (P Frame adjustment, without B Frame)

Video Bit-rate: 1Mbps~13Mbps each channel

Rate Control: CBR/VBR

Audio Encoding:

Audio format: MPEG1 Layer II, LC-AAC, HE-AAC and AC3 Pass through, support audio gain adjustment

Sampling rate: 48 KHz

Audio Bit-rate:

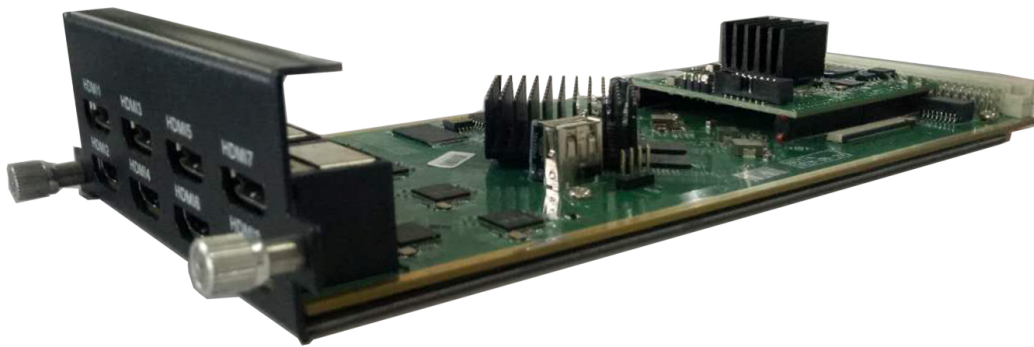
MPEG-1 Layer 2: 48/56/64/80/96/112/128/160/192/224/256/320/384 kbps LC-AAC:

48/56/64/80/96/112/128/160/192/224/256/320/384 kbps

HE-AAC: 48/56/64/80/96/112/128 kbps

Support Logo, Caption, QR Code insertion

DX228S-V2 8 HDMI Encoder Card



Input: 8*HDMI (4 HDMI is available)

Output: 8*SPTS (4 SPTS if 4 HDMI) output over UDP/RTP/RTSP, Unicast/Multicast

Video Encoding:

Video format: HEVC/H.265, MPEG-4 AVC/H.264

Input resolution: 1920×1080_60P, 1920×1080_60i, 1920×1080_50P, 1920×1080_50i, 1280×720_60P, 1280×720_50P, 720×576_50i, 720×480_60i,

Output resolution: 1920×1080_30P, 1920×1080_25P, 1280×720_30P, 1280×720_25P, 720×576_25P, 720×480_30P,

GOP structure: IP...P (P Frame adjustment, without B Frame)

Video Bit-rate: 1Mbps~13Mbps each channel

Rate Control: CBR/VBR

Audio Encoding:

Audio format: MPEG1 Layer II, LC-AAC, HE-AAC and AC3 Pass through, support audio gain adjustment

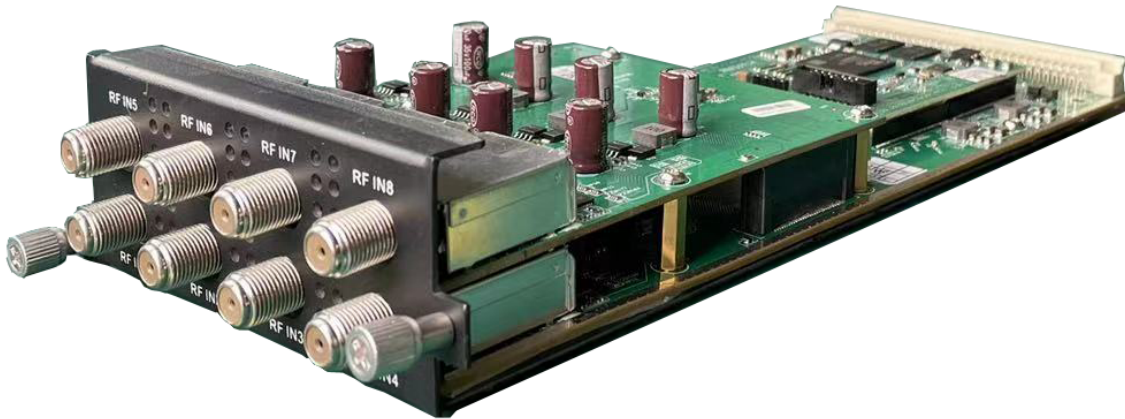
Sampling rate: 48 KHz

Audio Bit-rate:

MPEG-1 Layer 2: 48/56/64/80/96/112/128/160/192/224/256/320/384 kbps LC-AAC:
48/56/64/80/96/112/128/160/192/224/256/320/384 kbps
HE-AAC: 48/56/64/80/96/112/128 kbps

Support Logo, Caption, QR Code insertion

DX908 8 FTA DVB-S/S2/S2X Tuner Card



Stream in: 8 Tuner input, F Type,

Stream out: 512 SPTS out over UDP/RTP/RTSP, Unicast/Multicast

Tuner input: DVB-S/S2/S2X

Symbol rate: QPSK/8PSK/16APSK 0.5~45 Msps; 8APSK/32APSK: 0.5~40Msps

Input Frequency: 950-2150MHz

DVB-S Constellation: QPSK

FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8

DVB-S2 Constellation: QPSK/8PSK/16APSK/32APSK

FEC Demodulation:

QPSK: 1/2, 2/3, 3/4, 5/6, 3/5, 4/5, 8/9, 9/10

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10

32APSK: 3/4, 4/5, 5/6, 8/9, 9/10

DVB-S2X Constellation: QPSK/8PSK/8APSK/16APSK/32APSK

FEC Demodulation:

QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18

8APSK: 5/9-L, 26/45-L

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90

32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9, 9/10

Support Diseqc function

Multiplexing:

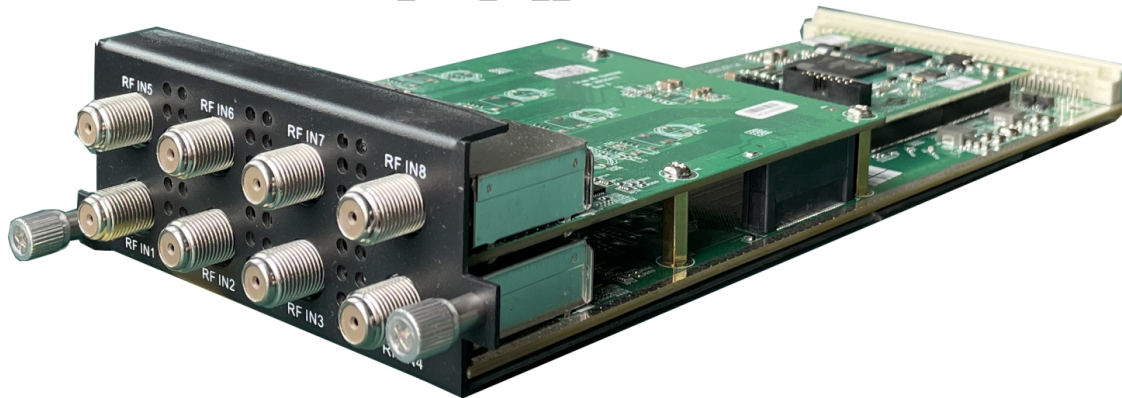
Maximum PID Remapping: 256 output pids

Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically

Descrambling:

BISS Mode: Mode 1, Mode E, up to 120Mbps, 32 BISS Key

DX928 8 FTA DVB-C/T/T2/ISDB-T Multi-Mode Tuner Card



Stream in: 8 tuner input, F Type

Stream out: 512 SPTS out over UDP/RTP/RTSP, Unicast/Multicast

Tuner input: DVB-C (J.83 A/C)/J.83B/ DVB-T/DVB-T2/ISDB-T switchable

Standard: DVB-C (J.83 A/C); J.83B

Input Frequency: 60MHz~890MHz

Symbol rate: 1000~9000Ksps

Constellation: 16/32/64/128/256 QAM; 64/256 QAM for J.83B

Standard: DVB-T/T2

Frequency In: 60MHz~890MHz

Bandwidth: 5/6/7/8M bandwidth

PLP Index: 0~255 for DVB-T2

Standard: ISDB-T

Input Frequency: 60-890MHz

Multiplexing:

Maximum PID Remapping: 256 output pids

Function: PID remapping (automatically or manually), generate PSI/ SI table automatically

Descrambling:

BISS Mode: Mode 1, Mode E; up to 120Mbps, 32 BISS Keys

Test data for reference (1U Model):

Protocol conversion	Programs	Bitrate	Terminals	CPU utilization
UDP to HTTP	60	8M	100	80%
UDP to HLS	60	8M	120	65%
UDP to HTTP	60	2M	400	80%
UDP to HLS	60	2M	400	40%
UDP to SRT	45	8M	---	80%
UDP to RTMP	50	8M	---	80%

Test data for reference (3U model):

Protocol conversion	Programs	Bitrate	Terminals	CPU utilization
UDP to HTTP	150	8M	600	82%
UDP to HLS	150	8M	1300	30%
UDP to SRT	130	8M	---	75%
UDP to RTMP	150	8M	---	80%

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More

than 20,000 television stations, radio stations, and hotel clients from different.

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